

PROGRAMMING FOR PROBLEM SOLVING

UNIT 1 C Fundamentals & Decision Making

PREMANAND S

Assistant Professor

Department of Computational Intelligence

SRM Institute of Science and Technology

Kattankulathur Campus

First Year Engineering

Q1: Data Type Detective

Question

Choose the **most appropriate data type** for each of the following:

- 1 Number of students in a class
- 2 Student name
- 3 CGPA
- 4 Registration number
- 5 Temperature
- 6 Whether the student passed or not
- 7 Number of books

Challenge

Give one reason for each of your choices.

Q1: Data Type Detective – Answer

Information	Data Type	Reason
Number of students	int	Whole number
Student name	char[]	Text / String
CGPA	double	Decimal value
Registration number	char[]	Contains letters and numbers
Temperature	double	Can contain decimals
Passed or not	_Bool	True / False
Number of books	int	Whole number

Key Point

A value containing digits is not necessarily an integer.

Q2: Find the Correct printf()

```
int age = 18;  
double cgpa = 8.75;  
char grade = 'A';  
  
printf("Age = %f\n", age);  
printf("CGPA = %d\n", cgpa);  
printf("Grade = %c\n", grade);
```

Task

- 1 Identify the **incorrect** printf() statements.
- 2 Correct the format specifiers.
- 3 Explain why each correction is required.

Q2: Find the Correct printf() – Answer

Corrected Program

```
int age = 18;
double cgpa = 8.75;
char grade = 'A';
printf("Age = %d\n", age);
printf("CGPA = %f\n", cgpa);
printf("Grade = %c\n", grade);
```

Incorrect Statements

- CGPA uses %d → Use %f for double.
- Age uses %f → Use %d for int.

Remember

int → %d

double → %f

char → %c

Q3: Predict the Output

```
int a = 10;  
int b = 3;  
printf("%d\n", a + b);  
printf("%d\n", a - b);  
printf("%d\n", a * b);  
printf("%d\n", a / b);  
printf("%d\n", a % b);
```

Task

- 1 Predict the output **without executing the program**.
- 2 Identify the operator used in each statement.
- 3 Why does `a / b` not produce a decimal value?

Think!

What is the difference between `/` and `%` when used with integers?

Q3: Predict the Output – Answer

Expected Output

13

7

30

3

1

Explanation

- $a + b \rightarrow 10 + 3 = 13$
- $a - b \rightarrow 10 - 3 = 7$
- $a * b \rightarrow 10 \times 3 = 30$
- $a / b \rightarrow 10/3 = 3$
- $a \% b \rightarrow 10\%3 = 1$

The & Symbol in scanf()

Remember

The symbol & is called the **address-of operator**.
It gives the **memory address** of a variable.

Expression	Meaning
age	Value of the variable
&age	Address of the variable

For Normal Variables

```
int age;  
double cgpa;  
char grade;  
scanf("%d", &age);  
scanf("%lf", &cgpa);  
scanf("%c", &grade);
```

Character Variable vs Character Array

Single Character

A single char variable needs &.

```
char grade;  
  
scanf("%c", &grade);
```

Character Array / String

A character array is used directly with scanf().

```
char name[50];  
  
scanf("%49s", name);
```

Character Variable vs Character Array

Easy Rule

Normal variable → **&**

Character array / string → **no &**

Q4: scanf() Detective

```
int age;  
  
printf("Enter age: ");  
scanf("%d", age);  
  
printf("Age = %d", age);
```

Task

- 1 Identify the mistake in the `scanf()` statement.
- 2 Correct the statement.
- 3 Why do we need `&` before `age`?

Think!

What does `scanf()` need in order to store the user's input?

Q4: scanf() Detective – Answer

Corrected Statement

```
int age;  
printf("Enter age: ");  
scanf("%d", &age);  
printf("Age = %d", age);
```

What Was the Mistake?

The original statement was:

```
scanf("%d", age);
```

It should be `scanf("%d", &age);`

Q5: String or Integer?

Question

Consider the following registration number:

26CSE1002J

Which is More Appropriate?

Option A: `int regno = 261002;`

Option B: `char regno[] = "26CSE1002J";`

Think!

- 1 Which option is more appropriate?
- 2 Why?
- 3 What about a phone number such as 0987654321?

Q5: String or Integer? – Answer

Answer

The appropriate data type depends on the **meaning and use** of the data.

Registration Number

```
char regno[] = "26CSE1002J";
```

Why? It contains both letters and digits and is used as an identifier.

Phone Number

```
char phone[] = "0987654321";
```

Why? A phone number is an identifier, not a quantity for arithmetic.

Key Point

Digits $\neq$ necessarily an integer

Use a data type based on **how the data is used**.

Q6: Find All Errors

```
#include <stdio.h>
int main()
{
    int age;
    double cgpa;
    printf("Enter age: ");
    scanf("%f", &age);
    printf("Enter CGPA: ");
    scanf("%d", &cgpa);
    printf("Age = %f\n", age);
    printf("CGPA = %d\n", cgpa);
    return 0;
}
```

Q6: Find All Errors – Answer

Corrected Program

```
#include <stdio.h>
int main()
{
    int age;
    double cgpa;
    printf("Enter age: ");
    scanf("%d", &age);
    printf("Enter CGPA: ");
    scanf("%lf", &cgpa);
    printf("Age = %d\n", age);
    printf("CGPA = %f\n", cgpa);
    return 0;
}
```

Q7: Compilation Error or Logical Error?

```
#include <stdio.h>
int main()
{
    int length = 10;
    int width = 5;
    int area = length + width;
    printf("Area = %d", area);
    return 0;
}
```

Q7: Compilation Error or Logical Error? – Answer

Logical Error

The program uses:

$$\text{area} = \text{length} + \text{width}$$

But the formula for the area of a rectangle is:

$$\text{Area} = \text{length} \times \text{width}$$

Therefore, the statement should be:

```
int area = length * width;
```

Key Point

A program can compile and run successfully, but still produce a wrong answer.

Q8: What Will Happen?

```
#include <stdio.h>
int main()
{
    int x = 10;
    printf("%d\n", x);
    scanf("%d", &x);
    printf("%d\n", x);
    return 0;
}
```

Input

Suppose the user enters:

25

Q8: What Will Happen? – Answer

Expected Output

10

25

Explanation

- 1 Initially, `x` is assigned the value 10.
- 2 The first `printf()` displays the current value of `x`.
- 3 `scanf()` reads the user's input 25.
- 4 The value 25 is stored in `x`.
- 5 The second `printf()` displays the updated value.

Key Point

`scanf()` **changes the value stored in the variable.**

Q9: Same Value, Different Data Type

```
int x = 10;  
double y = 10.0;  
printf("%d\n", x);  
printf("%.1f\n", y);
```

Task

- 1 Both x and y represent the value 10. Are they stored in the same way?
- 2 Why are different format specifiers used?
- 3 What is the difference between %d and %.1f?

Think!

Does the data type affect how a value is represented and displayed?

Q9: Same Value, Different Data Type – Answer

Answer

Although both variables represent the value 10, they have **different data types** and are stored differently.

Correct Output

```
10
10.0
```

Why Different Format Specifiers?

- x is an int → use %d
- y is a double → use %f

Key Point

The format specifier should match the data type of the variable.

Q10: Input or Output?

```
#include <stdio.h>
int main()
{
    int age;
    printf("Enter your age: ");
    scanf("%d", &age);
    printf("You are %d years old.\n", age);
    return 0;
}
```

Task

- 1 Which statement displays a message to the user?
- 2 Which statement receives input from the user?
- 3 What is the purpose of `&age`?
- 4 Which statement displays the value entered by the user?

Q10: Input or Output? – Answer

Answer

- 1 `printf("Enter your age: ");` → **Output statement**
- 2 `scanf("%d", &age);` → **Input statement**
- 3 `&age` → Gives the **memory address of** age where the input should be stored.
- 4 `printf("You are %d years old.", age);` → **Output statement** that displays the value entered by the user.

Remember

`printf()` → **Display / Output**

`scanf()` → **Read / Input**

Key Point

`scanf()` stores the user's input in a variable using its address.

Q11: Predict the Output

```
#include <stdio.h>
int main()
{
    int a = 10;
    double b = 2.5;
    printf("%d\n", a);
    printf("%.1f\n", b);
    printf("%.1f\n", a + b);
    return 0;
}
```

Task

- 1 Predict the output without executing the program.
- 2 What is the data type of $a + b$?
- 3 Why does $a + b$ produce a decimal value?

Q11: Predict the Output – Answer

Expected Output

10
2.5
12.5

Explanation

- a is an int with value 10.
- b is a double with value 2.5.
- In `a + b`, the int value is converted to double.
- Therefore:

$$10 + 2.5 = 12.5$$

Key Point

**When an int and a double are used together,
the int is converted to double.**

Q12: Find the Output

```
#include <stdio.h>
int main()
{
    int marks;
    printf("Enter marks: ");
    scanf("%d", &marks);
    printf("Marks = %d\n", marks);
    printf("Marks + 5 = %d\n", marks + 5);
    return 0;
}
```

Input

Suppose the user enters:

75

Q12: Find the Output – Answer

Expected Output

```
Marks = 75  
Marks + 5 = 80
```

Explanation

- The user enters 75.
- `scanf()` stores 75 in `marks`.
- `marks + 5` calculates:

$$75 + 5 = 80$$

- The expression `marks + 5` does **not** change the value of `marks`.

Key Point

An expression can calculate a new value without changing the original variable.

Q13: What Happens to the Original Value?

```
#include <stdio.h>
int main()
{
    int x = 20;
    int y;
    y = x + 10;
    printf("x = %d\n", x);
    printf("y = %d\n", y);
    return 0;
}
```

Task

- 1 Predict the output.
- 2 What values are stored in x and y?
- 3 Does the statement `y = x + 10;` change the value of x?

Q13: What Happens to the Original Value? – Answer

Expected Output

x = 20

y = 30

Explanation

- Initially, x contains the value 20.
- The expression $x + 10$ is evaluated:

$$20 + 10 = 30$$

- The result 30 is assigned to y.
- The value of x remains unchanged.

Key Point

**Using a variable in an expression does not change its value.
The value changes only when a new value is assigned to it.**

Q14: Assignment vs Comparison

```
#include <stdio.h>
int main()
{
    int age = 18;
    printf("Age = %d\n", age);
    age = 20;
    printf("Age = %d\n", age);
    return 0;
}
```

Task

- 1 Predict the output.
- 2 What happens to the original value 18?
- 3 What does `age = 20;` do?
- 4 Is `=` used to compare two values?

Q14: Assignment vs Comparison – Answer

Expected Output

Age = 18

Age = 20

Explanation

- Initially, age contains 18.
- The statement `age = 20;` assigns a new value to age.
- The previous value 18 is replaced by 20.
- Therefore, the second `printf()` displays 20.

Key Point

`=` → **Assignment**

`==` → **Comparison**

Q15: Spot the Mistake

```
#include <stdio.h>
int main()
{
    int mark1 = 80;
    int mark2 = 90;
    int average;
    average = (mark1 + mark2) / 2.0;
    printf("Average = %d\n", average);
    return 0; }
```

Task

- 1 Will the program compile?
- 2 Is the data type of average appropriate?
- 3 What will be the value printed?
- 4 How would you modify program to correctly store a decimal average?

Q15: Spot the Mistake – Answer

Answer

- The program will **compile successfully**.
- The expression

$$(80 + 90)/2.0 = 85.0$$

produces a double result.

- But average is declared as an `int`.
- Therefore, the decimal part is discarded when the result is stored.
- The program prints:

Average = 85

Q15: Spot the Mistake – Answer

Better Approach

Use a floating-point data type for the average:

```
double average;
```

Then:

```
average = (mark1 + mark2) / 2.0;
```

Key Point

Choose the data type based on the type of result you need.